DTU Compute
Department of Applied Mathematics and Computer Science

Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



Occupancy-Aware Climate Control
Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis
7 December, 2026

By
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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

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Signature

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Date

Abstract

ABSTRACT IS MISSING

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Mads Holme Håkonsson, MSc Civil Engineering, DTU
Creator of this thesis template.

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1 Introduction

Reduction in the usage of energy in buildings is a crucial aspect if one wants to go towards a more sustainable future. It has long been known that buildings are responseible for X% of the total energy consumption in the world, and that a large part of this energy is for HVAC systems [MISSING REFERENCE] HVAC systems energy consumption has been directly related to the occupant behaviours of the buildings. This imposes a challenge, as one has to be able to understand the behaviour of the occupants in the buildings before we can gain the largest optimization in energy consumption. (MISSING REFERENCE) has shown that there is a large variability in the behavour of occupants such as thermal comfort, lighting & ventiliation needs and precense hours. Within the context of energy

consumption in buildings, heating has been estimated to contribute to X% [MISSING REFERENCE] of the consumption. An aspect of heating of a building is the byproduct of the thermal comfort of the occupants in the building, and therefore, if we can understand the thermal comforts of the occupants in the buildings, we can optimize the energy consumption drasitically. However, the thermal comfort curves is a personal preference of the occupants, and we should therefore expect this to differer dependent on the individual buildings and the number of occupants.

This projects aims to understand when the occupants of a single offices are present, and how much thermal comfort feedback does we need to collect from the occupants in order to be able to infer the true thermal comfort of the occupants in the building.

WRITTEN OUTLINE OF EACH CHAPTER AND WHAT QUESTIONS THEY ANSWER

A Title

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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